number and configuration of hidden layers greatly influence the network's ability to learn complex patterns. Networks with more hidden layers are referred to as deep neural networks.

3. **Output Layer**: The output layer generates the final predictions or classifications of the neural network. The number of neurons in this layer depends on the specific task. For example, in a binary classification task, the output layer typically contains one neuron, while a multi-class classification task requires multiple neurons, each representing a class.

Key Concepts

- **Weights and Biases**: Weights represent the strength of the connection between two neurons, determining how much influence one neuron has on another. Biases are additional parameters that shift the activation function, enabling the network to fit the data more flexibly. Together, weights and biases form the trainable parameters of a neural network.
- **Activation Functions**: Activation functions introduce non-linearity into the network, allowing it to model complex relationships in the data. Common activation functions include:
    - **ReLU (Rectified Linear Unit)**: Sets all negative values to zero, preserving positive values. It is computationally efficient and widely used.
    - **Sigmoid**: Maps input values to a range between 0 and 1, often used in the output layer for binary classification.
    - **Tanh**: Similar to Sigmoid but maps values to a range between -1 and 1, providing stronger gradients for negative values.
- **Loss Function**: The loss function quantifies the difference between the predicted output and the actual target values. It provides feedback to the network during training, guiding it to adjust its parameters. Examples include Mean Squared Error